Uncertainty-Aware Colorization of Astronomical Images: A Monte Carlo Dropout Approach with Perceptual Validation

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Abstract

Composite color images of narrowband astronomical observations (the "Hubble Palette," mapping SII, Hα, and OIII filters to red, green, and blue channels) are produced by deterministic, hand-tuned stretches that give no indication of per-pixel confidence. We present a pipeline that instead learns this mapping with a compact convolutional neural network trained under simulated channel dropout and sensor noise, and that uses Monte Carlo Dropout at inference time to obtain, in addition to a color estimate, a pixel-wise uncertainty map. We apply the method to Hubble Space Telescope WFC3 narrowband observations of M16 (the Eagle Nebula / Pillars of Creation), retrieved from the Mikulski Archive for Space Telescopes (MAST). Qualitatively, the resulting uncertainty map is elevated at saturated stars, frame edges, and cosmic-ray-contaminated pixels, and low across the well-observed nebular body. We further validate the map with a web-based perceptual survey ($n = 9$ raters, 14 patches spanning the observed uncertainty range): while model uncertainty does not significantly correlate with the mean human "naturalness" rating, it correlates significantly with the degree of *disagreement among raters* (Spearman $\rho = 0.59$, $p = 0.03$), suggesting that regions the model flags as uncertain are also regions that are genuinely ambiguous to independent human observers. All code is released publicly to support reproduction and extension to other targets.

Keywords: astronomical image processing — deep learning — uncertainty quantification — Bayesian neural networks — Hubble Space Telescope

1 Introduction

Public-facing astronomical images are almost always composites: multiple monochrome exposures, typically in narrowband filters isolating specific emission lines (e.g., [S II] $\lambda 6716, 6731$,

Hα λ6563, [O III] λ5007), are individually stretched and assigned to color channels. The de facto standard for emission nebulae is the "Hubble Palette" (SII→R, Hα →G, OIII→B), popularized by the Hubble Heritage Project. While visually effective, this process is deterministic and offers no mechanism to communicate where the resulting color is well-constrained by the data and where it is not — for example, in regions where one band is missing, saturated, or contaminated by cosmic rays.

Bayesian deep learning offers a practical route to per-pixel confidence estimates without the computational cost of a full Bayesian neural network. Gal & Ghahramani (2016) showed that a network trained with Dropout can be interpreted as an approximation to a deep Gaussian process, and that keeping Dropout *active* at inference time and averaging multiple stochastic forward passes (Monte Carlo Dropout, MC-Dropout) yields both a predictive mean and a predictive variance, the latter serving as an uncertainty estimate. MC-Dropout has been applied broadly in medical imaging and autonomous driving, where it is important to know not just what a model predicts but how much to trust that prediction; its application to astronomical image colorization, to our knowledge, is not established in the mainstream literature, motivating the exploratory pipeline described here.

This paper makes three contributions: (1) an end-to-end, reproducible pipeline for MC-Dropout-based uncertainty-aware colorization of multi-band astronomical imagery, from raw MAST retrieval through registration, training, and tiled inference; (2) a qualitative analysis of the resulting uncertainty map on a real HST dataset of M16; and (3) the design (and preliminary results) of a perceptual survey intended to test whether the model's uncertainty is perceptually meaningful to human observers.

2 Data

We use narrowband imaging of M16 (NGC 6611, the Eagle Nebula) obtained with the Wide Field Camera 3 (WFC3/UVIS) aboard the Hubble Space Telescope, retrieved via `astroquery.mast` (Ginsburg et al., 2019; Astropy Collaboration, 2018). Three filters were used, each isolating a different ionic emission line:

- **F502N** ([O III], λ5007)
- **F656N** (Hα, λ6563)
- **F673N** ([S II], λ6716, 6731)

The OIII and SII exposures originate from the same observing proposal (proposal ID 13926) and share a common pointing. No Hα exposure from the same proposal was available in the archive; the closest-coverage Hα observation (proposal 16359) was retrieved instead. Because its footprint does not fully coincide with the OIII/SII pointing, the three bands were reprojected onto a common World Coordinate System (WCS) grid using `reproject.reproject_interp` (Robitaille, 2019), and the image was cropped to the largest axis-aligned rectangle with 100% valid coverage in all three bands, found via a maximal-rectangle-in-binary-matrix search over the union-validity mask. This crop is necessary because the naive bounding box of the valid-pixel mask still contains invalid corners when the input footprints are mutually rotated.

3 Methods

3.1 Baseline colorization

As a reference, we construct a deterministic Hubble Palette composite: each registered, cropped band is independently stretched with a 1st–99.5th percentile linear stretch and assigned to its corresponding RGB channel. This baseline has no notion of confidence and serves only as a visual and numerical reference point for the learned model.

3.2 MC-Dropout network

We use a compact fully convolutional network (four convolutional layers, 32–64 channels, ReLU activations, 1×1 output convolution with a sigmoid nonlinearity) with `Dropout2d` ($p = 0.3$) inserted after each hidden layer. Because no independently verified "true color" exists for these data, we cannot train a supervised colorization model in the conventional sense. Instead we frame the learning problem as *reconstruction under simulated data degradation*: during training, each input patch has, with probability 0.4, one of its three bands set to zero (simulating a missing/uncovered band, as encountered with the real Hα footprint mismatch), plus additive Gaussian noise ($\sigma = 0.05$) applied to all channels. The training target is the clean, undegraded three-band patch. This framing gives the resulting uncertainty a concrete operational meaning: it should be elevated wherever the network is extrapolating from incomplete or noisy information, which is directly relevant to the missing-coverage problem observed in Section 2.

The network is trained with Adam (learning rate 10^{-3}) and an MSE loss on 128×128 patches randomly sampled from the registered, percentile stretched bands, for 30 epochs (4000 patches/epoch, batch size 16), on a single NVIDIA RTX 4060 laptop GPU.

3.3 Uncertainty inference

At inference, Dropout is kept active (`model.train()` mode) and the full image is processed in overlapping 256×256 tiles (stride 192) to avoid seam artifacts. Each tile is passed through the network $N = 30$ times; the per-pixel mean across passes is taken as the color estimate and the per-pixel standard deviation as the uncertainty estimate. Tiles are recombined by averaging in overlap regions.

3.4 Perceptual validation survey

To assess whether the resulting uncertainty map corresponds to human judgments of color trustworthiness, we selected 14 180×180-pixel patches spanning the observed range of mean per-patch uncertainty (from the lowest to the highest decile), with display order randomized per session. Human raters, blind to the underlying uncertainty values, scored each patch on a 5-point scale (1 = "completely artificial," 5 = "completely natural and trustworthy"). The instrument is a self-contained, browser-based survey (Supplementary material; `survey/uncertainty_survey.html`) that requires no server-side infrastructure. The version shown to raters was in Persian; the file in the repository is an English translation of

the same interface. We planned to correlate the mean human rating per patch against the model's mean uncertainty for that patch using Spearman's rank correlation coefficient ρ.

4 Results

4.1 Registration and baseline

After reprojection, the OIII and SII bands (originating from the same pointing) retained 93% pixel coverage relative to the reference frame, while the independently-pointed Hα band retained only 26–33% coverage depending on the specific archival exposure tested. The maximal-rectangle crop yields a final working image of 1920×3873 pixels with 100% valid coverage in all three bands.

4.2 Uncertainty map

Qualitatively, the MC-Dropout uncertainty map (Figure 1) shows three consistent patterns: (i) elevated uncertainty at a saturated foreground star, (ii) elevated uncertainty along image borders, where the network has reduced spatial context, and (iii) elevated uncertainty at scattered single-pixel and short-streak features consistent with cosmic-ray hits, which by construction appear in only one of the three independently-exposed bands and therefore constitute exactly the kind of "missing/inconsistent information" the training procedure was designed to flag. The bulk of the nebular body, where all three bands are well exposed and mutually consistent, shows low uncertainty.

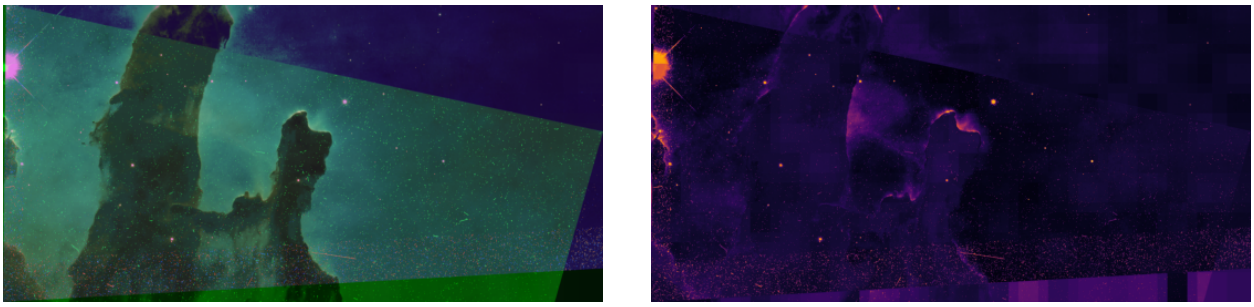


Figure 1: Left: mean MC-Dropout color prediction for a representative region of M16. Right: corresponding per-pixel uncertainty map (standard deviation across $N = 30$ stochastic forward passes), with brighter colors indicating higher uncertainty.

4.3 Perceptual survey results

Nine raters completed the survey ($n = 9$; one rater skipped a single patch, giving $n = 8$ for that item). Two complementary correlations were tested against the model's mean per-patch uncertainty using Spearman's rank correlation coefficient: (i) the mean human "naturalness" rating per patch, and (ii) the standard deviation of ratings across raters for that patch, i.e., the degree of *inter-rater disagreement*.

The correlation between model uncertainty and mean human rating was not statistically significant ($\rho = -0.227$, $p = 0.435$): patches with higher model uncertainty were not consistently rated as less natural on average. However, the correlation between model uncertainty and *inter-rater disagreement* was positive and statistically significant ($\rho = 0.591$, $p = 0.026$, $n = 14$ patches). That is, patches on which the model was more uncertain were also patches on which human raters disagreed more strongly with one another about naturalness (Figure 2). The patch with the single highest model uncertainty (containing a saturated foreground star, roughly 3× the uncertainty of any other patch) also had among the highest inter-rater standard deviations in the entire set.

We consider this a more informative and more robust signal than the mean-rating correlation for a dataset of this size: the mean rating of a single rater is a noisy point estimate of an inherently subjective quantity, whereas disagreement *among* several independent raters is a direct, model-free proxy for how ambiguous a patch actually is — which is conceptually close to what an epistemic uncertainty estimate is intended to capture. The result suggests that regions the network flags as uncertain correspond to regions that are also genuinely ambiguous to independent human observers, even though it does not yet establish that the model's uncertainty predicts the *direction* (more vs. less trustworthy) of that ambiguity.

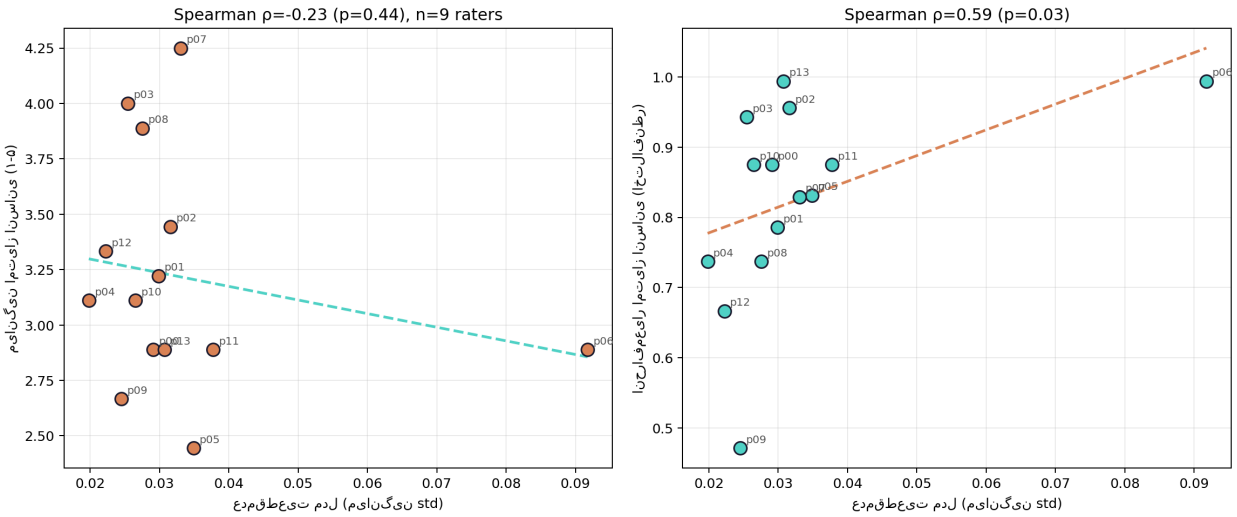


Figure 2: Left: mean human naturalness rating (1–5) vs. model uncertainty per patch; the correlation is not significant ($\rho = -0.23$, $p = 0.44$). Right: standard deviation of human ratings (inter-rater disagreement) vs. model uncertainty per patch; the correlation is significant ($\rho = 0.59$, $p = 0.03$). $n = 9$ raters, 14 patches.

We note two caveats. First, with $n = 9$ raters and 14 patches, this result should be treated as an initial finding rather than a definitive validation; we plan to continue data collection toward $n \geq 20$ raters to obtain a more stable estimate of both effects and to test whether the mean-rating correlation strengthens with a larger, less noisy sample. Second, the disagreement metric is itself noisy at $n = 9$ per patch; a pre-registered replication with a larger, independent rater pool would strengthen this result considerably.

5 Discussion

Several limitations should be noted. First, because no ground-truth color exists, the network's training target is itself a deterministic stretch of the observed bands rather than an independently verified "correct" color; the uncertainty we estimate is therefore best understood as *data-completeness/consistency uncertainty* (how much the network relies on extrapolation from degraded input) rather than a calibrated statistical confidence interval on true physical flux. Second, while the disagreement-based correlation reported in Section 4.3 is statistically significant, it is based on only 9 raters and 14 patches from a single target; the mean-rating correlation remains non-significant at this sample size, and both results should be treated as preliminary pending a larger, pre-registered replication. Third, results are demonstrated on a single target (M16); generalization to other nebulae, filter sets, and instruments (e.g., JWST NIRCam) remains to be tested.

6 Conclusion and Future Work

We have demonstrated a complete, reproducible pipeline for MC-Dropout-based uncertainty-aware colorization of multi-band Hubble Space Telescope imagery, including automated data retrieval, robust registration under mismatched fields of view, a degradation-aware training scheme that gives the resulting uncertainty a concrete operational interpretation, and a perceptual survey instrument for human validation. In an initial survey of 9 raters, model uncertainty did not significantly predict mean human naturalness ratings, but did significantly predict *inter-rater disagreement* ($\rho = 0.59$, $p = 0.03$), suggesting the model's uncertainty estimate captures genuine perceptual ambiguity. Immediate next steps are (1) expanding the survey toward $n \geq 20$ raters to test whether this result strengthens and whether the mean-rating effect becomes detectable, (2) extending the pipeline to additional targets and instruments, and (3) exploring alternative uncertainty-estimation techniques (deep ensembles, evidential deep learning) as a comparison point for MC-Dropout. All code, the survey instrument, and this manuscript source are released publicly (see repository).

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During the preparation of this work, the author used Claude (Anthropic), a large language model, to assist with drafting and editing the manuscript text, writing and debugging the analysis code, analyzing the survey results, and translating code comments, figure labels, and the survey interface into English. The author reviewed and edited the AI-assisted output and takes full responsibility for the content of this work.

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